Electricidad I:
Impedancia y admitancia

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Impendacia

Impedancia

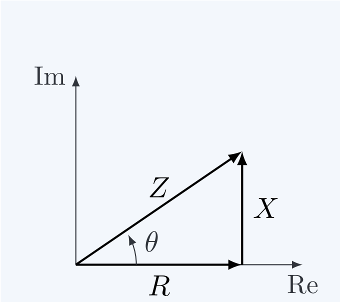
$$\boldsymbol{Z} = \frac{\boldsymbol{v}}{\boldsymbol{i}}$$

$$\boldsymbol{Z} = R + jX = |\boldsymbol{Z}|\angle\theta$$

$$|\boldsymbol{Z}| = \sqrt{R^2 + X^2} \qquad \theta = \arctan\left(\frac{X}{R}\right)$$

$$R = |\boldsymbol{Z}|\cos\theta \qquad X = |\boldsymbol{Z}|\operatorname{sen}\theta$$

donde R es la resistencia y X es la reactancia, ambas en ohms (Ω)



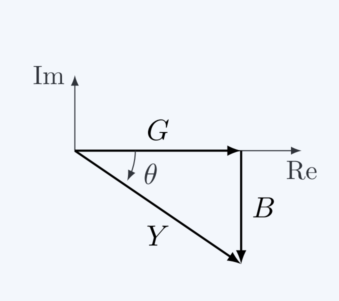
Admitancia

Admitancia

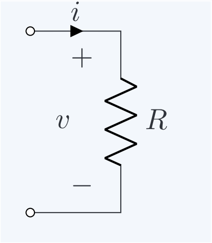
$$\boldsymbol{Y} = \frac{\boldsymbol{i}}{\boldsymbol{v}}$$

$$\boldsymbol{Y} = G + jB = \frac{1}{|\boldsymbol{Z}|}\angle - \theta$$

donde G es la conductancia y B es la susceptancia, ambas en siemens (S)



Circuito resistivo



$$i = I \,\mathrm{sen}\,(\omega t + \phi)$$

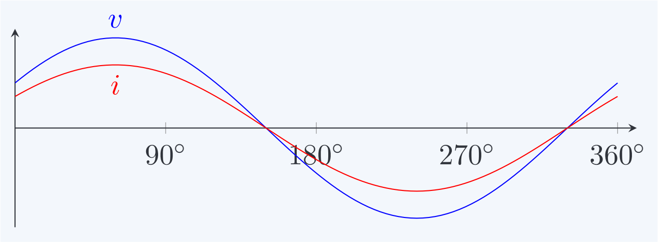
$$v = iR = RI \,\mathrm{sen}\,(\omega t + \phi)$$

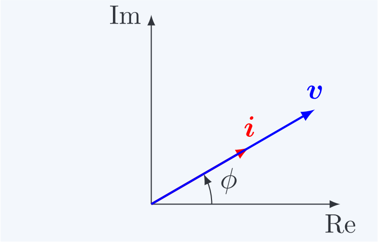
$$\boldsymbol{i} = I\angle\phi$$

$$\boldsymbol{v} = R\boldsymbol{i} = RI\angle\phi$$

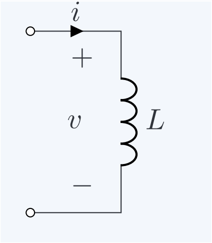
$$\boldsymbol{Z} = \frac{\boldsymbol{v}}{\boldsymbol{i}}$$

$$\boldsymbol{Z} = R + j0 = R\angle 0^\circ$$





Circuito inductivo



$$i = I\,\mathrm{sen}\,(\omega t + \phi)$$

$$v = L\frac{di}{dt} = \omega L I\,\mathrm{sen}\,(\omega t + \phi + 90^\circ)$$

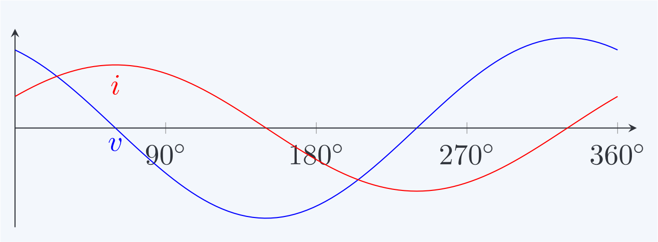
$$\boldsymbol{i} = I\angle\phi$$

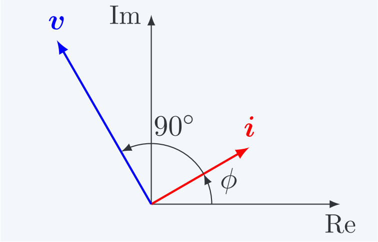
$$\boldsymbol{v} = j\omega L\boldsymbol{i} = \omega L I\angle\phi + 90^\circ$$

$$\boldsymbol{Z} = \frac{\boldsymbol{v}}{\boldsymbol{i}}$$

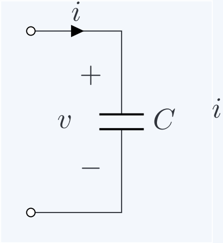
$$\boldsymbol{Z} = 0 + j\omega L = \omega L\angle 90^\circ$$

$$X_L = \omega L$$





Circuito capacitivo



$$v = V \operatorname{sen}(\omega t + \phi)$$

$$i = C\frac{dv}{dt} = \omega C V \operatorname{sen}(\omega t + \phi + 90^\circ)$$

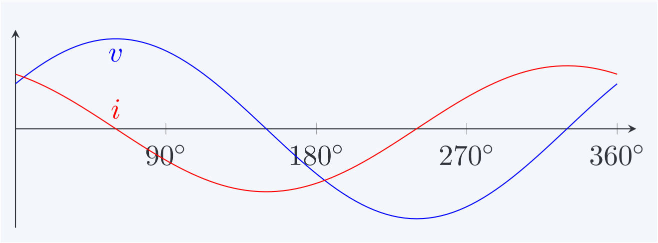
$$\boldsymbol{v} = V\angle\phi$$

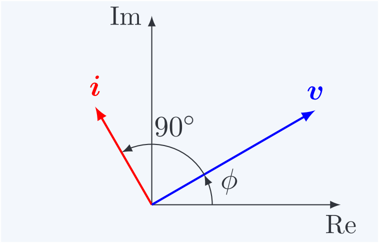
$$\boldsymbol{i} = j\omega C\boldsymbol{v} = \omega C V\angle\phi + 90^\circ$$

$$\boldsymbol{Z} = \frac{\boldsymbol{v}}{\boldsymbol{i}}$$

$$\boldsymbol{Z} = 0 + \frac{1}{j\omega C} = \frac{1}{\omega C}\angle - 90^\circ$$

$$X_C = \frac{1}{\omega C}$$





Impedancias en serie

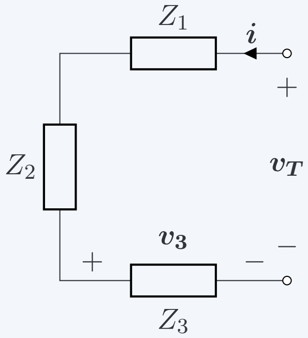
Impedancia equivalente

$$Z_{eq} = Z_1 + Z_2 + Z_3$$

Divisor de voltaje

$$\boldsymbol{v_n} = \boldsymbol{v_T} \cdot \frac{Z_n}{Z_1 + Z_2 + \cdots + Z_n}$$

$$\boldsymbol{v_3} = \boldsymbol{v_T} \cdot \frac{Z_3}{Z_1 + Z_2 + Z_3}$$



Impedancias en paralelo

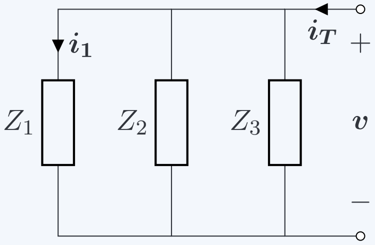
Impedancia equivalente

$$\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}$$

Divisor de corriente

$$\boldsymbol{i_n} = \boldsymbol{i_T} \cdot \frac{\frac{1}{Z_n}}{\frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_n}}$$

$$\boldsymbol{i_1} = \boldsymbol{i_T} \cdot \frac{\frac{1}{Z_1}}{\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}}$$



Impedancias en paralelo: dos impedancias

Impedancia equivalente

$$Z_{eq} = \frac{Z_1 \cdot Z_2}{Z_1 + Z_2}$$

Divisor de corriente

$$i_1 = i_T \cdot \frac{Z_2}{Z_1 + Z_2}$$